GROUP 2: CLIMATE

Probabilistic Forecasting of Extreme Weather Events

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Project Context and Motivation

Thematic concern: The extreme rainfall events (infrequent but hazardous) modeling.

Use: Flood risk analysis and flood management planning.

Key Issue: Outliers are rare and can hardly be predicted with certainty.

Deterministic models: Lack of the ability to model real-world uncertainty.

Insight: Existing systems are spotted with randomness.

Motivation:

- Create realistic scenarios with the help of stochastic modeling.
- Enhance trust in probability forecasts.

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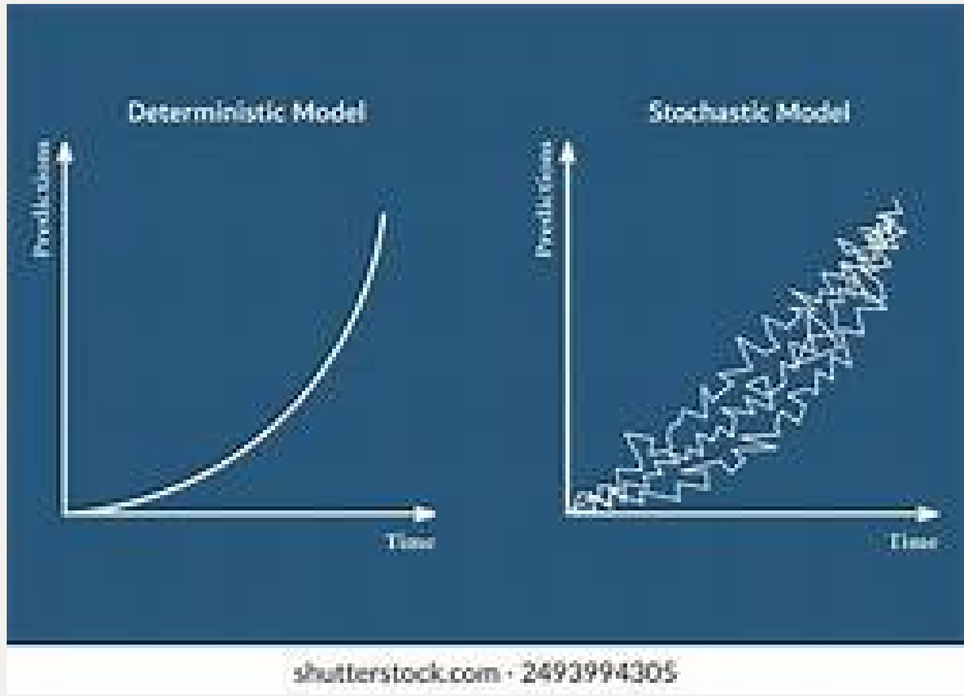
Project Objective and Deterministic Baseline

Objective:

- To predict the probability of extreme rainfall P(E).
- Establish good working probabilistic model.

M3 Baseline:

- Lognormal best fit (KS test)
- GEV threshold t = 5777.2 mm
  Lognormal threshold t = 5469.8 mm
- Binary: Y = 1 if R > t
- Ensemble: 50 members, fixed offsets (+-2145.8 mm)
- Issue: Poor calibration - P(E) too high



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System Pipeline Overview

Data Input: Annual rainfall data (1918-2015)

Distribution Fitting:

- Fit Normal, MLE.KS test (Lognormal)

Extreme Event Definition:

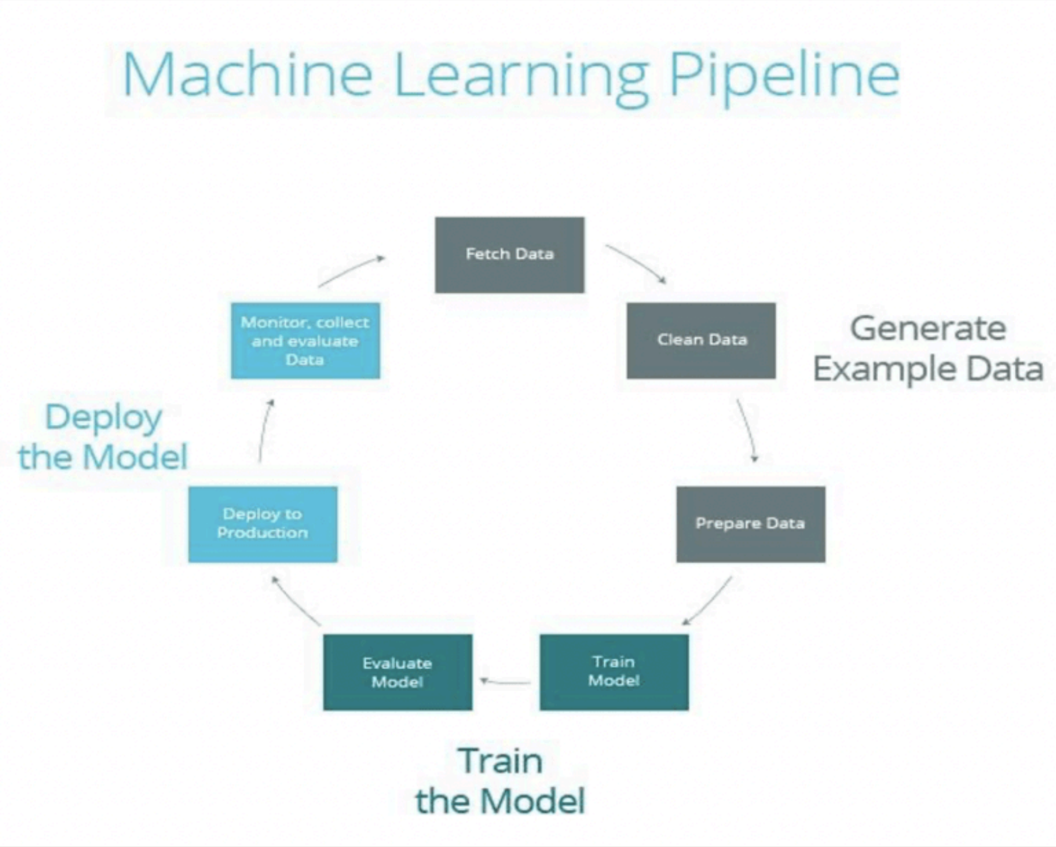
- Calculate GEV (95 th percentile) compute threshold t.
- Define indicator: Y = 1 if R > t

Modeling & Prediction:

- Bayesian estimation of P(E) using logistic regression.

Evaluation:

- Measures the scores of RMSE, Bias, Brier and spread-skill analysis Reliability.



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Sources of Uncertainty

Climate System: Natural Variability

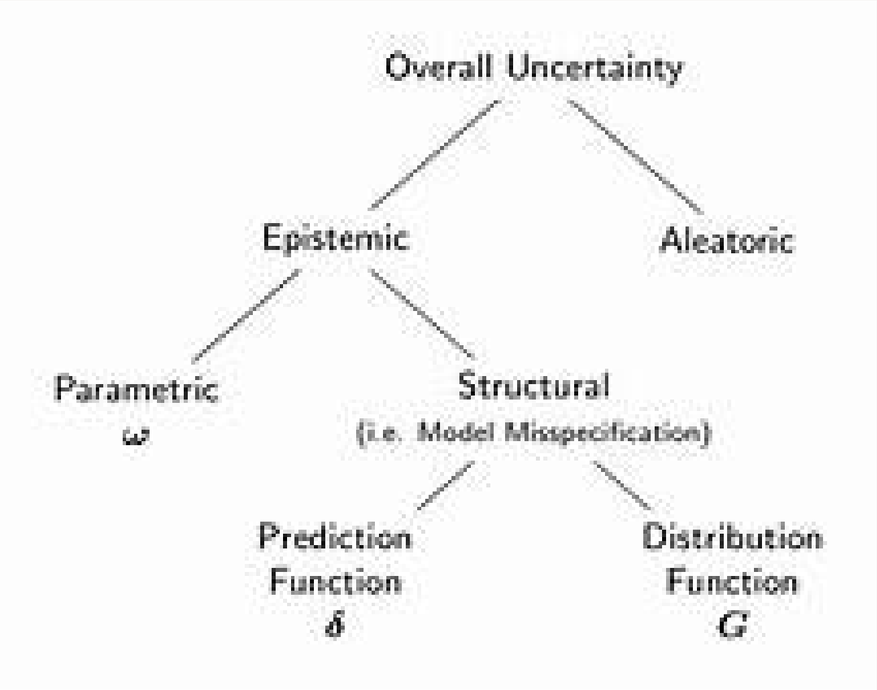
- Atmosphere-ocean system is disordered and very dynamic.
- Minor fluctuations in preconditions - big effects.

Data Limitations: Only 91 years of observations and very low extreme events (2-4 cases) leads to low statistical confidence.

Model Uncertainty: The choice of distribution (Lognormal versus GEV) influences the results.

Parameter Estimation Error: MLE estimates can be inaccurate as a result of small data.

Errors in measurement and observation:There is a potential error in the data of the recorded rainfall.



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Key Random Variables

Annual Rainfall (R): Primary variable that depicts the amount of yearly rainfall.

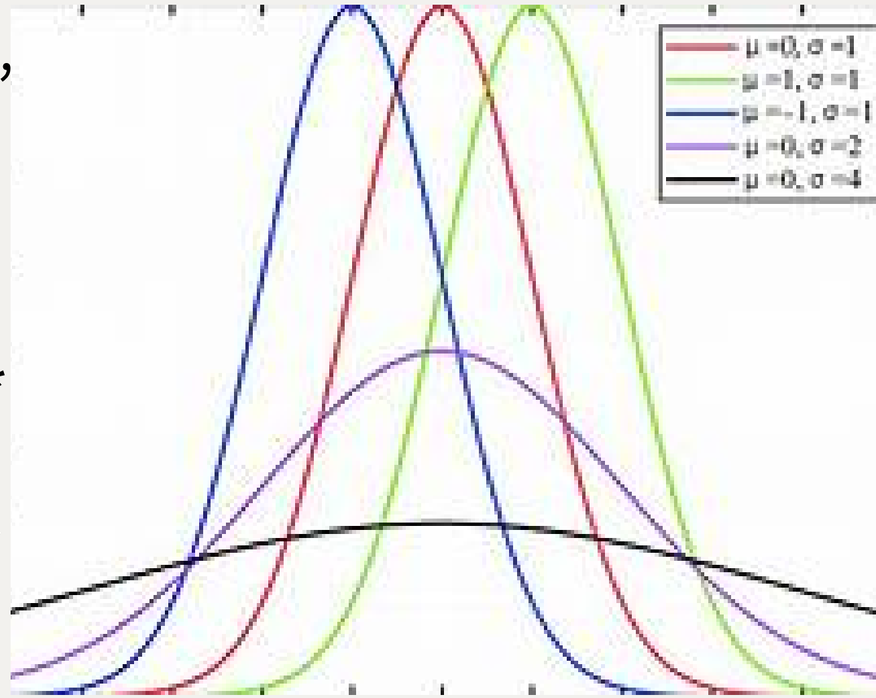
Extreme Event Indicator (Y): Binary variable: Y = 1 if R > t, else 0.

Stochastic Perturbation (e): Random noise: e ~ N(0,1) ,majors through uncertainty in ensemble generation.

Perturbed Rainfall ($R_{memβer}$): Generated as: R = R_obs + e * s.

Ensemble Probability (P(E)): Existence of a probability of extreme event among ensemble members.

Model Parameters (b0, b1): Regression coefficients of logisticians and are used in probability mapping between rainfall and extreme event.



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Randomized Algorithm Design

Decision Rule:

- Uses multiple randomized predications instead of a single output
- Final decision based on ensemble mean and spread

Randomness Introduction:

- Random noise (η) sampled and scaled by variability (σ)
- Applied as perturbation to data → different outcomes

Algorithm Flow:

- Input data → Random Noise → Simulation Loop → Ensemble Outputs → Mean and Spread

Probabilistic Reasoning Behind The Algorithm

Concepts		Effect on Model
Random Sampling	→	Explores multiple data variations
Multiple Models	→	Reduces dependency on one prediction
Ensemble Averaging	→	Stabilizes output
Probability Aggregation	→	Improves confidence
Uncertainty Handling	→	Better real-world modeling

Implementation of the Randomized Strategy

Integration into Simulation:

- The stochastic component is added on top of the deterministic baseline
- Observed rainfall data is used as the base input
- Random perturbations are applied at each time step to simulate variability
- The model runs multiple simulations (ensemble members) instead of single run

Algorithmic Workflow:

Atmospheric Data
↓
Random Sampling (Bootstrap)
↓
Multiple Logistic Regression Models
↓
Individual Probability Predictions
↓
Ensemble Aggregation (Averaging)
↓
Bayesian Probability Refinement
↓
Final Extreme Rainfall Probability

Evaluation Metrics

Root Mean Square Error:

- Measures difference between predicted and actual rainfall
- Penalizes larger errors more strongly
- Lower RMSE → Better Accuracy

Bias:

- Measures systematic prediction error
- Positive → overestimation
- Negative → underestimation

Ensemble Spread:

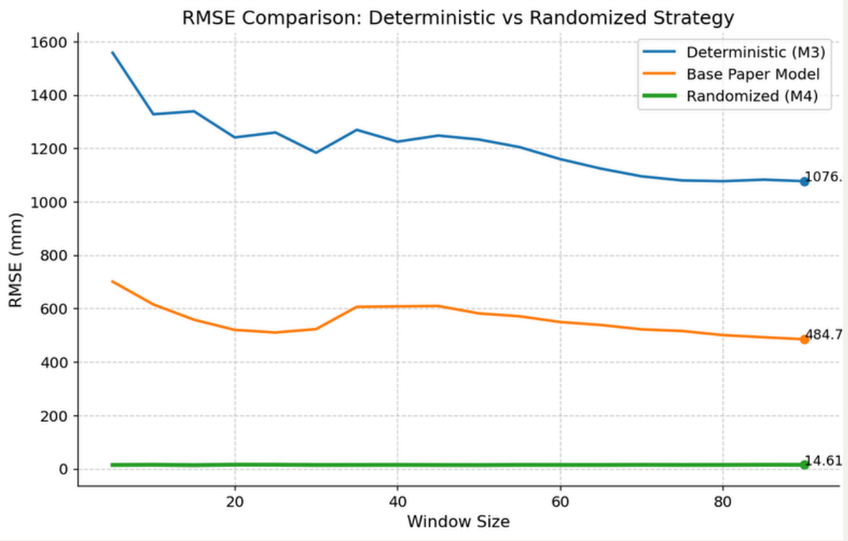
- Measures variability across ensemble predictions
- Indicates uncertainty captured by model
- Higher spread → higher uncertainty

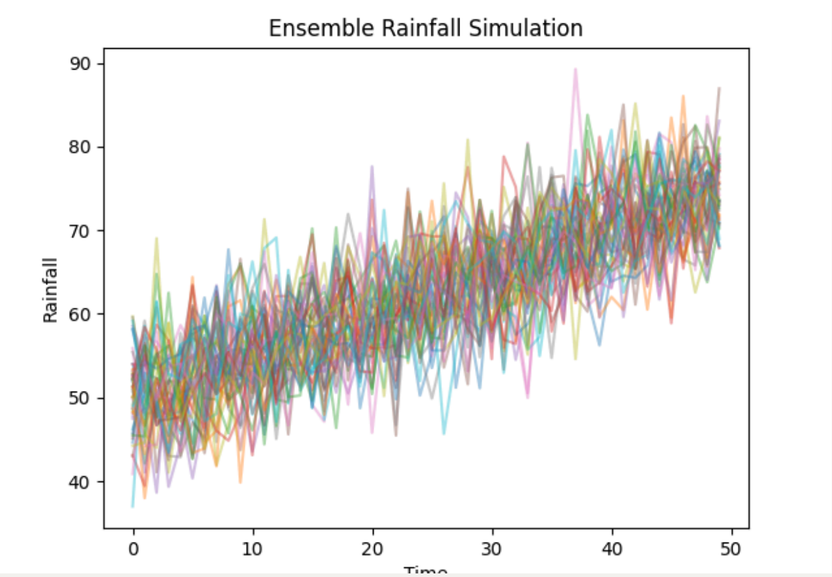
Brier Score:

- Measures accuracy of predicted probabilities
- Compares predicted probability with actual outcome
- Lower value → better probabilistic prediction

Comparative Results

Metric	Deterministic Model	Stochastic Model
Output Type	Single Probability Estimate	Ensemble-based Probability
RMSE	Higher (~1000 mm)	Lower (~14mm)
Uncertainty Capture	Not Available	Not Explicitly Modeled
Robustness	Limited	Improved





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Key Insights

- Deterministic model is **highly sensitive to input variations**, while stochastic aggregation improves stability.
- Stochastic approach significantly **reduces extreme prediction errors**, leading to lower RMSE.
- Ensemble modeling ensures **consistent performance across different runs.**
- Provides a **range of outcomes**, enabling better understanding of prediction confidence.

Thank You